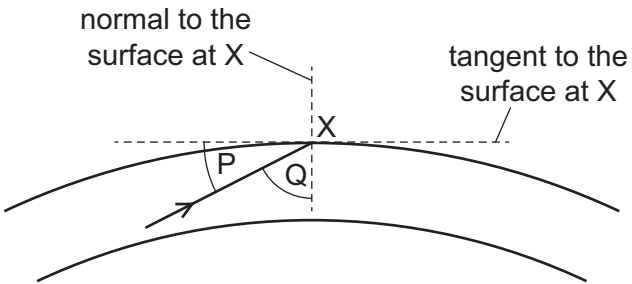
24 The diagram shows a ray of light inside an optical fibre approaching point X.



The light is totally reflected within the fibre.

The refractive index of the material of the optical fibre is 1.39.

Which condition ensures that total internal reflection occurs?

A $Q < 45°$ **B** $Q > 47°$ **C** $P > 47°$ **D** $P = Q$